Exercise Sheet 6

Social Context

5942UE Network Science

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6.A Homophily and Measurement

Exercise 6.A.1: Measuring Homophily by Hand

A friendship network of 8 students: 4 study CS (C_1–C_4) and 4 study Business (B_1–B_4).

Edges: C_1-C_2, C_1-C_3, C_2-C_3, C_3-C_4, C_4-B_1, B_1-B_2, B_2-B_3, B_3-B_4.

1. Count edges within CS, within Business, and across departments.
2. Compute the **E-I index**: $(E - I)/(E + I)$, ranging from -1 (pure homophily) to $+1$ (pure heterophily).
3. Is this network homophilic, heterophilic, or neutral with respect to department?
4. What would the E-I index be if ties formed **randomly** regardless of department?

Exercise 6.A.2: Homophily vs. Confounding

Using `nx.karate_club_graph()` with the `"club"` attribute as the group label:

1. Count within-faction and cross-faction edges.
2. Compute the **E-I index** for faction.
3. Compute the expected cross-faction edge fraction under **random mixing**.
4. Is the network homophilic? By how much relative to random?

6.B Affiliation and Similarity

Exercise 6.B.1: Three Closure Types

Affiliation network: students S_1, S_2, S_3, S_4 and courses C_1, C_2, C_3.

Enrolments: S_1–C_1, S_1–C_2, S_2–C_1, S_3–C_2, S_3–C_3, S_4–C_3.

1. Project onto the student layer (two students connected if they share a course).
2. Identify one example each of: (a) **triadic closure**, (b) **focal closure**, (c) **membership closure**.
3. If S_2 joins C_2, does this enable any new closure?
4. What information is **lost** when projecting?

Exercise 6.B.2: Bipartite Projection in Python

Implement the affiliation network from 6.B.1 in NetworkX and compute a weighted projection onto the student layer.

6.C Social Dynamics and Causal Inference

Exercise 6.C.1: Distinguishing Social Forces

For each scenario, state whether it is best explained by **selection**, **socialization**, or **confounding**:

1. Two students both like jazz, become friends, and a year later both discover classical music.
2. Two colleagues at a tech startup both use Python; their friendship forms day 1.
3. Students in the same dormitory become friends and all prefer the same pizza place.
4. A smoker quits after befriending non-smokers.

For each: what data (cross-sectional vs. longitudinal, observational vs. experimental) is needed to identify the mechanism?

Exercise 6.C.2: Designing a Study

You want to know whether students in the same study group become more similar in performance (socialization) or whether similar students form study groups together (selection).

1. What cross-sectional data would you collect?
2. What longitudinal data is needed to separate the mechanisms?
3. What would an ideal experimental design look like?
4. What would you observe if **both** mechanisms operate?

6.D Spatial Segregation

Exercise 6.D.1: Schelling Model Analysis

In Schelling's model, agents on a grid prefer at least k of 8 neighbours to be the same type. Dissatisfied agents move.

1. If $k = 1$, do you expect strong segregation? Why?
2. If $k = 5$ (majority), what outcome do you predict?
3. Why is it surprising that mild preferences ($k = 3$ or $k = 4$) can still produce strong global segregation?
4. How does this connect to homophily? What is the "network" in Schelling's model?

Exercise 6.D.2: Schelling Simulation

Implement a **1D Schelling model**: agents in a line; each needs at least `threshold` fraction of left+right neighbours to be the same type. Unhappy agents swap with random unhappy agents of the opposite type.

1. Implement and run for 100 steps.
2. Visualise initial and final states.
3. Vary `threshold` from 0.1 to 0.9 and plot average run-length at convergence.
4. At what threshold does strong segregation emerge?

Wrap-Up

- homophily is measurable; the E-I index is the basic diagnostic
- bipartite affiliations explain a lot of structure that pure friendship graphs hide
- selection, socialization, and confounding cannot be separated from cross-sectional data alone
- mild local preferences can still produce sharp global segregation